

JAMES RUSE AGRICULTURAL HIGH SCHOOL
YEAR 11 TERM 4 ASSESSMENT
4 UNIT 1999

Time: 85 minutes

QUESTION 1 : START A NEW PAGE (6 MARKS)

(a) Given $z_1 = 3 + i$ and $z_2 = 4 - 2i$

find (i) $|z_1|$

(ii) $\bar{z}_2$

(iii) $\frac{z_1}{z_2}$ in the form $a + bi$

(b) Graph on the Argand diagram the position vectors z_1 , $\bar{z}_2$ and z_3 where $z_3 = \frac{z_1}{z_2}$

QUESTION 2 : START A NEW PAGE (14 MARKS)

(i) Solve the equation $z^3 = 1$ for the roots P_0 , P_1 and P_2 expressed in both mod - arg form and the form $a + ib$.

(ii) Show these roots on the Argand diagram clearly indicating their relative positions.

(iii) Deduce that if ω is a non - real cube root of unity then $1 + \omega + \omega^2 = 0$

(iv) Given $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ and if $e = a + b$,

$f = a\omega + b\omega^2$ and $g = a\omega^2 + b\omega$ where ω is a non - real cube root of unity

simplify $e^3 + f^3 + g^3$ in terms of a and b .

(v) Given $(2 - 3i)^3 = -46 - 9i$ find another value of z such that $z^3 = -46 - 9i$

QUESTION 3 : START A NEW PAGE (20 Marks)

(a) (i) Graph the region: $|z - 4 + i| \leq 1$

(ii) Graph the locus: $\text{Arg}(z - 3i) = \frac{-\pi}{4}$

(b) A complex root of $z^2 + (3 - i)z + k = 0$ is $2 + i$.

Find (i) the value of the other root

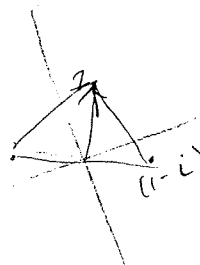
(ii) the value of k

(c) Given that $z^2 = 8 - 6i$ (i) find the values of z

(ii) show that one of the values of z is a solution to the equation

$$z^4 - 15z^2 - 6z + 110 = 0$$

(d) If $\frac{z - 1 + i}{z + 1 - i} = ki$ where $k \in \mathbb{R}$ find with the aid of a diagram the value of $|z|$, giving reasons.



QUESTION 4: START A NEW PAGE (20 MARKS)

(a) Describe the locus of :

(i) $|z| = |z - 3|$

(ii) $\text{Arg}(z - 3) - \text{Arg}(z + 3) = \frac{\pi}{2}$

(iii) What value of z satisfies both (i) and (ii) ?

(b) Simplify $(\sqrt{3} + i)^{12} + (\sqrt{3} - i)^{12}$

(c) If $z = \cos \theta + i \sin \theta$ then

(i) show that $\cos n\theta = \frac{z^n + z^{-n}}{2}$ and $\sin n\theta = \frac{z^n - z^{-n}}{2i}$

(ii) find an expression in z as a fraction in simplest terms for

$$1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta$$

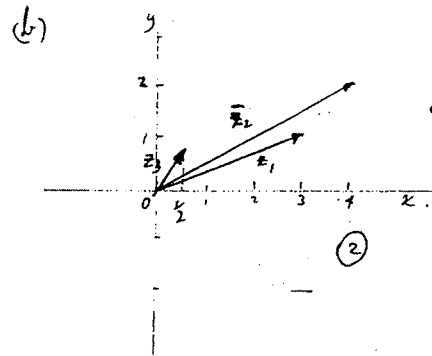
(iii) find an expression in z for
$$\frac{\cos \frac{n\theta}{2} \sin \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}$$

hence deduce a relationship between (ii) and (iii) .

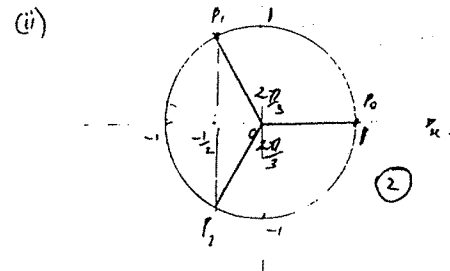
END OF EXAM



1 (a) (i) $|z_1| = \sqrt{3^2 + 1^2} = \sqrt{10} \text{ units}$ (1)
 (ii) $\bar{z}_2 = 4 + 2i$ (1)
 (iii) $\frac{z_1}{z_2} = \frac{3+i}{4-2i} \cdot \frac{4+2i}{4+2i}$
 $= \frac{12+7i-2}{16+4}$
 $= \frac{10+7i}{20}$
 $= \frac{1}{2} + \frac{7}{20}i$ (2)



2 (i) $z^3 = 1$
 $z^3 = 1 [\cos 2\pi k + i \sin 2\pi k]$
 $z = \cos \frac{2\pi k}{3} + i \sin \frac{2\pi k}{3}$
 $P_0 = \cos 0 = 1 + 0i$ (1)
 $k=1 \quad P_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$
 $= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ (2)
 $k=2 \quad P_2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$
 $= -\frac{1}{2} - \frac{\sqrt{3}}{2}i$ (2)



(iii) $w^3 = 1$
 $w^3 - 1 = 0$
 $(w-1)(w^2+w+1) = 0$
 If w is not real $w \neq 1$ (1)
 $\therefore w$ satisfies $w^2 + w + 1 = 0$
 $g^3 = a^3w^6 + 3a^2bw^5 + 3ab^2w^4 + b^3$
 $= a^3 + 3a^2bw^2 + 3ab^2w + b^3$
 $f^3 = a^3w^3 + 3a^2bw^4 + 3ab^2w^5 + b^3$
 $= a^3 + 3a^2bw + 3ab^2w^2 + b^3$
 $\therefore e^3 + f^3 + g^3 = a^3 + 3a^2b + 3ab^2 + b^3 + a^3 + 3a^2bw + 3ab^2w^2 + b^3 + a^3 + 3a^2bw^2 + 3ab^2w + b^3$
 $= 3a^3 + 3b^3 + 3[a^2b][1+w+w^2] + 3ab^2[1+w+w^2]$
 $= 3[a^3 + b^3]$ (3)

(iv) $f = aw + bw^2 \quad g = aw^2 + bw$
 $\therefore f^3 = a^3w^3 + 3a^2bw^4 + 3ab^2w^5 + b^3$
 $= a^3 + 3a^2bw + 3ab^2w^2 + b^3$
 $\therefore e^3 + f^3 + g^3 = a^3 + 3a^2b + 3ab^2 + b^3 + a^3 + 3a^2bw + 3ab^2w^2 + b^3 + a^3 + 3a^2bw^2 + 3ab^2w + b^3$
 $= 3a^3 + 3b^3 + 3[a^2b][1+w+w^2] + 3ab^2[1+w+w^2]$
 $= 3[a^3 + b^3]$ (3)

(v) $z = (2-3i) \left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i \right)$
 $z_1 = \frac{3\sqrt{3}}{2} - 1 + \left(\sqrt{3} + \frac{3}{2} \right)i$ OR $z_2 = -1 - \frac{3\sqrt{3}}{2} + \left(\frac{3}{2} - \sqrt{3} \right)i$ (2)

3(a) (i)

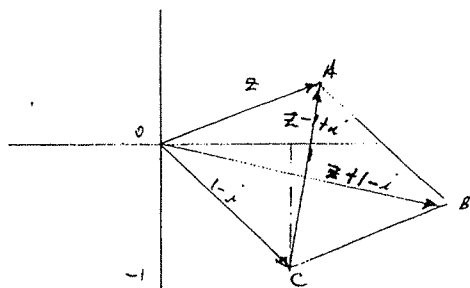
(ii)

(b) (i) $z + 2 + i = -3 + i$
 other root $z = -5$ (2)
 (ii) $k = -5(2+i)$
 $= -10 - 5i$ (2)

(c) (i) $z^2 = 8 - 6i$
 let $z = x + iy \quad x, y \in \mathbb{R}$
 $(x+iy)^2 = 8 - 6i$
 $x^2 - y^2 + 2xyi = 8 - 6i$
 $\therefore x^2 - y^2 = 8$
 $2xy = -6 \Rightarrow y = -\frac{3}{x}$
 $\therefore x^2 - \left(-\frac{3}{x} \right)^2 = 8$
 $x^4 - 8x^2 - 9 = 0$
 $(x^2 - 9)(x^2 + 1) = 0$
 $x = 3$ And $x = -3$ And $x^2 + 1 = 0$ has no real solution $x \in \mathbb{R}$
 $y = -1$ $y = 1$
 $\therefore z_1 = 3 - i \quad z_2 = -3 + i$ (4)

(ii) $P(z) = z^4 - 15z^2 - 6z + 110$
 $P(3-i) = (8-6i)^2 - 15(8-6i) - 6(3-i) + 110$
 $= 64 - 96i - 36 - 120 + 90i - 18 + 6i + 110$
 $= 0$
 $P(-3+i) = (8-6i)^2 - 15(8-6i) - 6(-3+i) + 110$
 $= 64 - 96i - 36 - 120 + 90i + 18 - 6i + 110$
 $= 0$
 $\therefore z_1 = 3 - i$ and $z_2 = -3 + i$ are the other roots.

(d)



$$\frac{z-1+i}{z+1-i} = k u$$

$$z-1+i = (z+1-i) k u$$

i vector $z-1+i \perp$ vector $z+1-i$ and modulus changed.

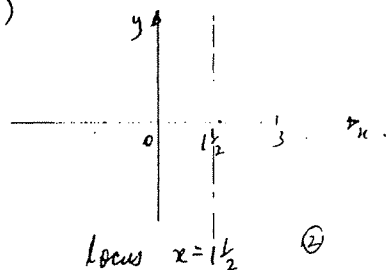
ii Diagonals OACB are at right angles in parallelogram OACB

∴ OACB is a rhombus

$$\therefore |z| = |k u|$$

$$= \sqrt{2} \text{ units.}$$

4(a) (i)



Locus $x = 1/2$ (2)

(ii)

$$z = 3 \cos \frac{\pi}{3}$$

$$= 3 \left[\frac{1}{2} + \frac{\sqrt{3}}{2} i \right]$$

$$= \frac{3}{2} + \frac{3\sqrt{3}}{2} i \quad (2)$$

$$(b) (\sqrt{3}+i)^{12} + (\sqrt{3}-i)^{12} = \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^{12} + \left[2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right) \right]^{12}$$

$$= 2^{12} \left(\cos 2\pi + i \sin 2\pi \right) + 2^{12} \left(\cos 2\pi - i \sin 2\pi \right)$$

$$= 2^{12}$$

$$= 8192 \quad (3)$$

(c)

$$z = \cos \theta + i \sin \theta$$

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$z^n = \cos n\theta + i \sin n\theta$$

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

$$\cos n\theta = \frac{z^n + z^{-n}}{2}$$

$$z^{-n} = (\cos \theta + i \sin \theta)^{-n}$$

$$= \cos(-n\theta) + i \sin(-n\theta)$$

$$z^{-n} = \cos n\theta - i \sin n\theta$$

$$\text{And } \frac{z^n - z^{-n}}{2} = i \sin n\theta$$

$$\sin n\theta = \frac{z^n - z^{-n}}{2i}$$

$$(ii) 1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta$$

$$= 1 + \frac{z+z^{-1}}{2} + \frac{z^2+z^{-2}}{2} + \dots + \frac{z^n+z^{-n}}{2}$$

$$= 1 + \frac{1}{2} \left[z + z^{-1} + \dots + z^n + z^{-n} \right] + \frac{1}{2} \left[z^{-1} + z^{-2} + \dots + z^{-n} \right]$$

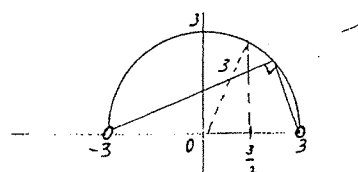
$$= 1 + \frac{z}{2} \left(\frac{z^n - 1}{z - 1} \right) + \frac{z^{-1}}{2} \left(\frac{z^{-n} - 1}{z^{-1} - 1} \right)$$

$$= 1 + \frac{z}{2} \left(\frac{z^n - 1}{z - 1} \right) + \frac{z^{-n} - 1}{2(1 - z)}$$

$$= \frac{2z - 2 + z^{n+1} - z - z^{-n} + 1}{2(z - 1)}$$

$$= \frac{z^{n+1} - z^{-n} + z - 1}{2(z - 1)} \quad (4)$$

(iv)



$$\text{Locus } x^2 + y^2 = 9$$

$$\{0 \leq y \leq 3\}$$

$$\{-3 \leq x \leq 3\} \quad (2)$$

$$\begin{aligned}
 (iii) \quad \frac{\cos \frac{n\theta}{2} \cdot \rho \sin \left(\frac{n+1}{2} \theta \right)}{\rho \sin \frac{\theta}{2}} &= \frac{z^{\frac{n}{2}} + z^{-\frac{n}{2}}}{2} \cdot \frac{z^{\frac{n+1}{2}} - z^{-\frac{n+1}{2}}}{2z} \\
 &= \frac{z^{\frac{n}{2}} + z^{-\frac{n}{2}}}{2} \cdot \frac{z^{\frac{n+1}{2}} - z^{-\frac{n+1}{2}}}{2z} \\
 &= \frac{z^{\frac{n+1}{2}} - 1 + z - z^{-n}}{2(z-1)} \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \therefore 1 + \cos \theta + \cos 2\theta + \cos 3\theta &= \frac{\cos \frac{\theta}{2} \cdot \rho \sin \left(\frac{n+1}{2} \theta \right)}{\rho \sin \frac{\theta}{2}} \\
 \sum_{k=0}^n \cos k\theta &= \frac{\cos \frac{\theta}{2} \cdot \rho \sin \left(\frac{n+1}{2} \theta \right)}{\rho \sin \frac{\theta}{2}} \quad (1)
 \end{aligned}$$

